Yarish Khan

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Professional Summary

- B.Tech (ECE) student building production AI systems with two deployed LLM applications featuring RAG pipelines, automated evaluation suites, and LLM observability.
- Solved **400+ LeetCode problems** with a strong foundation in Data Structures, Algorithms, backend engineering, and scalable AI application development.

Education

Jamia Millia Islamia – New Delhi Expected May 2027

Bachelor of Technology (B.Tech), Electronics and Communication Engineering

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Technical Skills

- **Languages:** C++, C, Python, JavaScript, SQL
- **AI Engineering:** Retrieval-Augmented Generation (RAG), Vector Embeddings, Similarity Search, LLM-as-Judge Evaluations, LLM Observability, Prompt Versioning, OpenRouter, Gemini APIs
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
- **Frontend:** React.js, React Native, Vite
- **Databases:** MongoDB Atlas, MySQL
- **Tools:** Git, GitHub, GitHub Actions (CI/CD), Docker, Linux, Cursor, GitHub Copilot

Projects

AI Pronunciation Assessment System Live Demo | GitHub

FastAPI, React, Whisper, Wav2Vec2, Phonemizer, OpenRouter, FFmpeg, Docker

- ***Architecture:*** Engineered an explainable pronunciation assessment pipeline that performs deterministic speech analysis before LLM coaching, separating objective scoring from generated explanations to reduce hallucinations.
- ***Pipeline:*** Built a modular backend integrating audio normalization, Whisper transcription, grapheme-to-phoneme conversion, whole-utterance phoneme recognition, IPA alignment, and automated report generation.
- ***Rule Engine:*** Developed a configurable pronunciation analysis engine that detects phoneme substitutions, insertions, deletions, and consonant articulation errors.
- ***Impact:*** Reduced false-positive pronunciation errors by approximately **80%** on clean recordings through IPA normalization, confidence gating, and function-word filtering.

Speakzy — AI Language Learning Platform Live Demo | GitHub

Node.js, Express, MongoDB Atlas, React, OpenRouter, Gemini Embeddings, Stream, Vitest

- ***RAG Personalization:*** Built retrieval-augmented flashcard generation grounded in study history using 768-dimensional Gemini embeddings and cosine similarity to reinforce previously learned vocabulary without repetition.
- ***LLM Evaluation:*** Developed a two-tier CI evaluation suite with a 16-case multilingual golden dataset and cross-family LLM-as-Judge evaluation, uncovering two critical production bugs during the first execution.
- ***Observability:*** Centralized every LLM request through a unified gateway to track latency, token usage, estimated cost, and cache status in a dedicated monitoring dashboard.
- ***Product Features:*** Developed a citation-grounded grammar tutor over an ingested Tatoeba corpus, a zero-LLM-cost spaced repetition system, and real-time chat/video using Stream.
- ***Security & DevOps:*** Secured the platform with 82 Vitest/Supertest integration tests running in GitHub Actions CI, tiered API rate limiting, and Zod validation for both user requests and LLM outputs.

Competitive Programming

- Solved **400+ LeetCode problems** covering dynamic programming, graphs, trees, heaps, greedy algorithms, binary search, and advanced data structure techniques.